

Project Specifications

Detecting AI-Generated Text Using ML and NLP Techniques

GitHub: <https://github.com/SabirShah1402/Detecting-AI-Generated-Text-Project>

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1.1 Project Topic

This project builds a machine learning and NLP system to classify text as human-written or AI-generated. As large language models (LLMs) become increasingly capable, reliable detection of AI content is critical for combating academic dishonesty, misinformation, and AI content misuse. The system extracts and analyses linguistic, stylometric, and semantic features from two real-world labelled corpora.

1.2 Research Question

Can machine learning and NLP models reliably distinguish AI-generated text from human-written text across multiple domains, and which combination of feature engineering and model architecture produces the most generalisable detection system?

1.3 Objectives

- **Data Engineering:** Load, clean, and merge the HC3 corpus (HuggingFace) and the Kaggle AI-Generated Text Detection dataset into a unified, reproducible preprocessing pipeline with stratified splits.
- **Feature Engineering:** Extract TF-IDF representations (unigram + bigram), stylometric features (word count, avg sentence length, punctuation density, type-token ratio), and token embeddings for neural models.
- **Model Development:** Train and evaluate three architectures:
 - (a) TF-IDF + Logistic Regression baseline
 - (b) custom Hybrid CNN (dual-stream: stylometric features + GloVe embeddings)
 - (c) fine-tuned RoBERTa-base transformer.
- **Optimisation:** Apply GridSearchCV (LR), manual grid search (CNN), and HuggingFace TrainingArguments sweep (RoBERTa) to maximise macro F1-score on the validation set.
- **Generalisation Testing:** Evaluate cross-dataset performance: train on HC3, test on Kaggle dataset and vice versa.
- **Deployment:** Deliver a Streamlit web application providing real-time classification with confidence scores, top-contributing features, and stylometric summaries for user-submitted text.

1.4 Goal / Target

Achieve at least 90% macro F1-score on the held-out test set with the best model, and demonstrate a functional Streamlit application that classifies arbitrary text in under 3 seconds. The system must generalise across writing styles and domains, not merely perform well within the training distribution.

1.5 Key Performance Indicators

KPI Metric	Min. Target	Good Target	How Measured
Macro F1-Score	≥ 0.83	≥ 0.90	Held-out test set
Overall Accuracy	$\geq 85\%$	$\geq 92\%$	Held-out test set
ROC-AUC Score	≥ 0.88	≥ 0.95	Held-out test set
Precision (AI class)	≥ 0.85	≥ 0.92	Test set - AI class only
Cross-dataset F1	≥ 0.75	≥ 0.85	Generalisation test
App Response Time	< 3 sec	< 1 sec	Measured during UAT

1.6 Datasets

- **HC3 — Human ChatGPT Comparison Corpus:** <https://huggingface.co/datasets/Hello-SimpleAI/HC3>
24,300 rows across five domains: Reddit ELI5, Finance, Medicine, Open QA, Wiki CSAI. Each row holds a list of human_answers and chatgpt_answers to the same question, plus a source field.
- **Kaggle AI-Generated Text Detection:** <https://www.kaggle.com/code/rsuhara/ai-generated-text-detection-quick-baseline>
Quick-baseline notebook by rsuhara using a labelled dataset with text and generated (0 = human, 1 = AI) columns. Covers diverse general-domain writing styles produced by multiple LLMs. Complements HC3 by providing out-of-domain text for cross-dataset generalisation testing.

1.7 Model Overview

- **Model 1 — Logistic Regression (baseline):** TF-IDF (1-2 grams, max 50k features) input. L2 regularisation, calibrated probabilities. Fast, interpretable lower-bound benchmark.
- **Model 2 — Hybrid CNN (custom neural network):** Dual-stream architecture: Stream A processes stylometric features through a Dense layer; Stream B processes GloVe-100d word embeddings through Conv1D → GlobalMaxPooling → Dense. Streams concatenate, then Dropout → sigmoid output.
- **Model 3 — RoBERTa-base (transfer learning):** roberta-base (125M parameters, HuggingFace) with a classification head. Fine-tuned end-to-end using the HuggingFace Trainer API with AdamW optimiser, linear warmup, and early stopping on val F1.

1.8 Application

A Streamlit web app accepts user-pasted text and returns: classification label (Human / AI-Generated), confidence percentage, top-5 TF-IDF terms contributing to the decision, and stylometric stats. Target users include educators, journalists, and content-moderation teams.